

SIGNAL

User Operations Manual

AI-Powered Energy Intelligence Platform

Shell Energy · Enterprise Crude Oil Market Intelligence

Version 2.0 · Analyst & Executive Edition

Classification: INTERNAL / CONFIDENTIAL

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1 System Overview

1.1 What SIGNAL Does

SIGNAL (System for Intelligent Geopolitical News and Asset Linking) is Shell Energy's enterprise-grade crude oil market intelligence platform. It combines real-time geopolitical news ingestion, natural language processing (NLP) sentiment analysis, and two distinct machine-learning models to deliver actionable price-direction and price-level signals to traders, analysts, and executives.

SIGNAL is a **decision-support system**, not an autonomous trading engine. It surfaces data-driven intelligence to augment human judgment — enabling faster, more informed responses to market-moving events.

1.2 Dual-Engine Architecture

Engine	Type	Output	Accuracy / Error
Direction Signal Engine	Classification (Random Forest + XGBoost)	UP ▲ or DOWN ▼ + confidence probability	81.5% overall accuracy Precision ~82% Recall ~80%
Price Forecast Engine	Regression (Stacking Ensemble: Ridge + HistGBM + LightGBM → RidgeCV)	Next-day WTI price in \$/bbl	RMSE: \$2.07/bbl R ² : 0.953 MAPE: ~2.5%

Key Principle:

Both engines run daily after market close. Results are published to the dashboard the following morning, giving users overnight intelligence before trading opens.

2 Getting Started

2.1 Login Process

Navigate to the SIGNAL URL in your browser. The login page is the default landing screen. Enter your registered email address and password, then click **Sign In**.

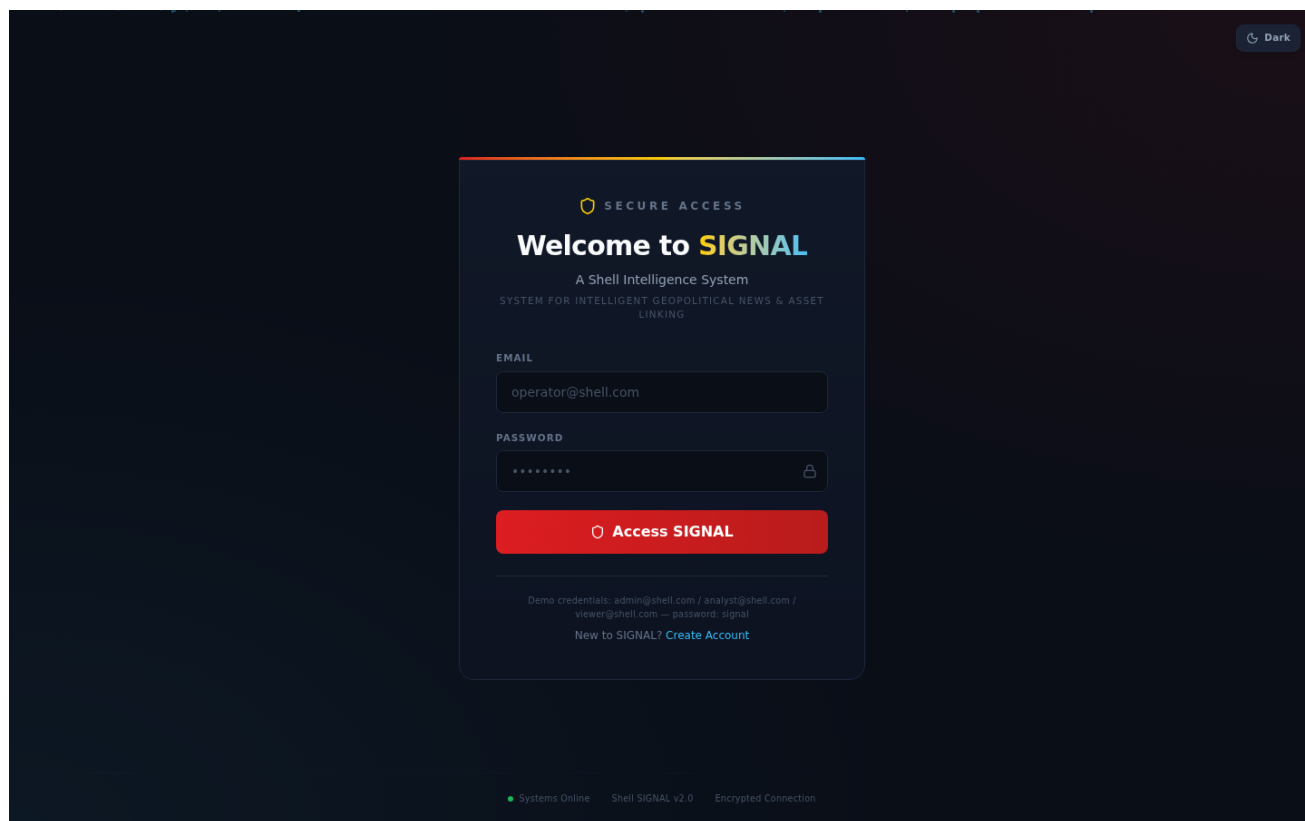


Figure 2-1 — SIGNAL Login Screen

2.2 Registration

New users click **Create an Account** on the login page. On the registration screen:

- Enter your full name and email address.
- Choose a secure password.
- Select your role: **Analyst** or **Executive**.
- Click **Create Account** — you will be redirected to the onboarding flow.

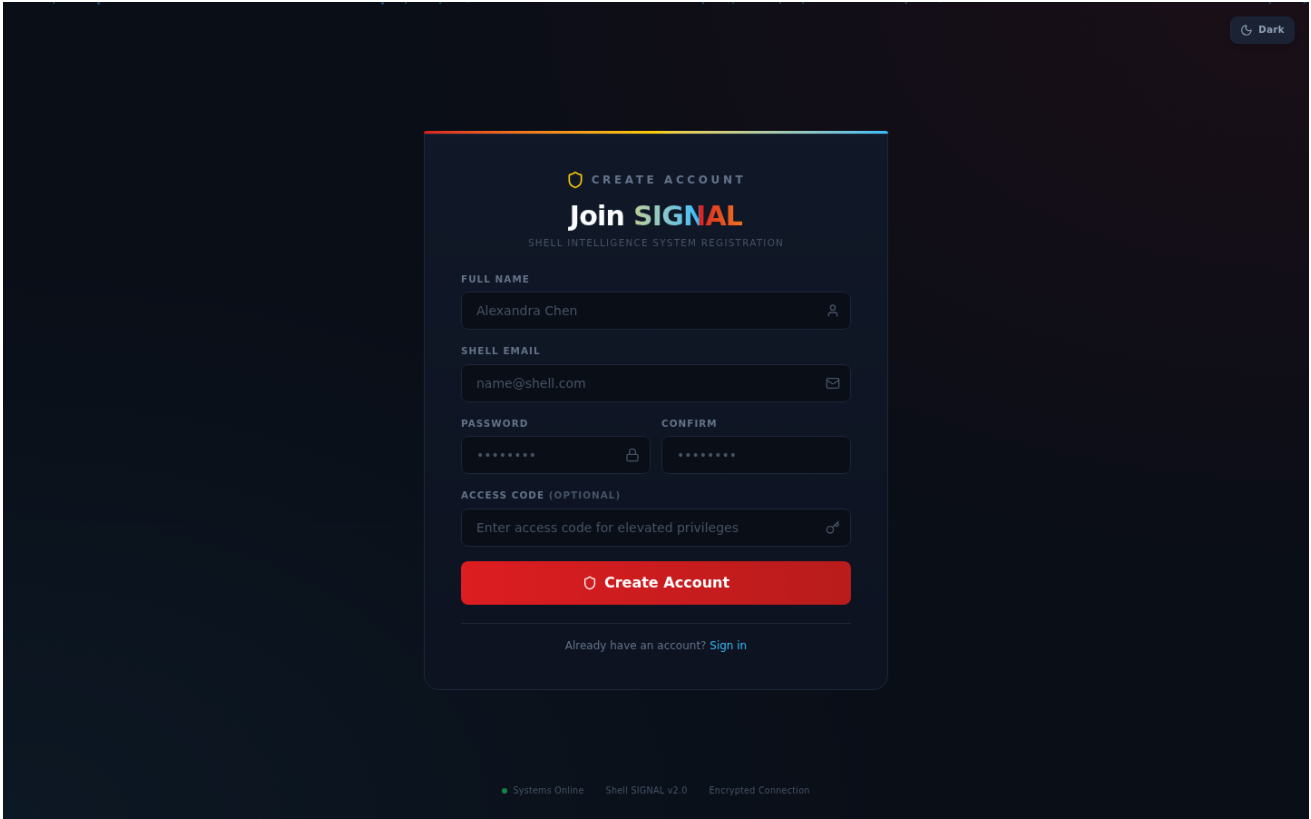


Figure 2-2 — SIGNAL Registration Screen

2.3 Role-Based Access

Role	Available Pages	Special Features
Analyst	Command Center, Intelligence Feed, Commodity View, Analytics Dashboard, Price Forecast, Settings	Standard badge in sidebar; Basic briefing flow; C
Executive	All Analyst pages + executive onboarding briefing	Gold EXEC badge in sidebar; Strategic briefing flow; C

2.4 Onboarding Flow

All new users complete a guided onboarding before accessing the full dashboard. The sequence differs by role:

Step	Analyst (5 Steps)	Executive (6 Steps)
1	Welcome & platform overview	Cinematic security verification (Phase 0)
2	Alert threshold configuration	Strategic briefing with live KPI data
3	Data source acknowledgement	Strategic focus area selection
4	Dashboard preview	Escalation protocols (cadence + level)
5	Activation — access granted	Dashboard preview
6	—	Activation — EXEC badge applied

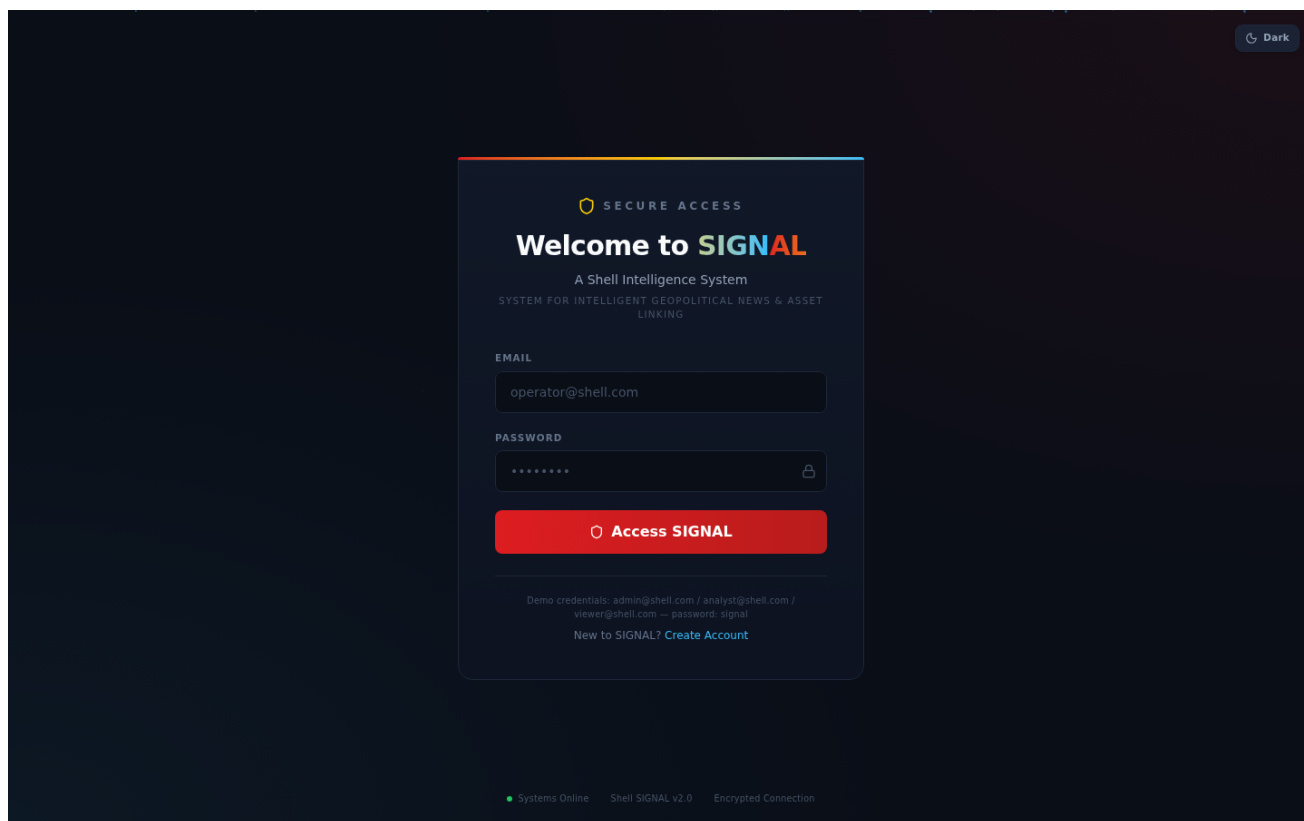


Figure 2-3 — Onboarding Flow (initial screen)

3 Dashboard Walkthrough

The Command Center (/dashboard) is the primary intelligence screen. It delivers a complete situational picture on a single view — current AI signal, live price, model performance, news flow, and alerts.

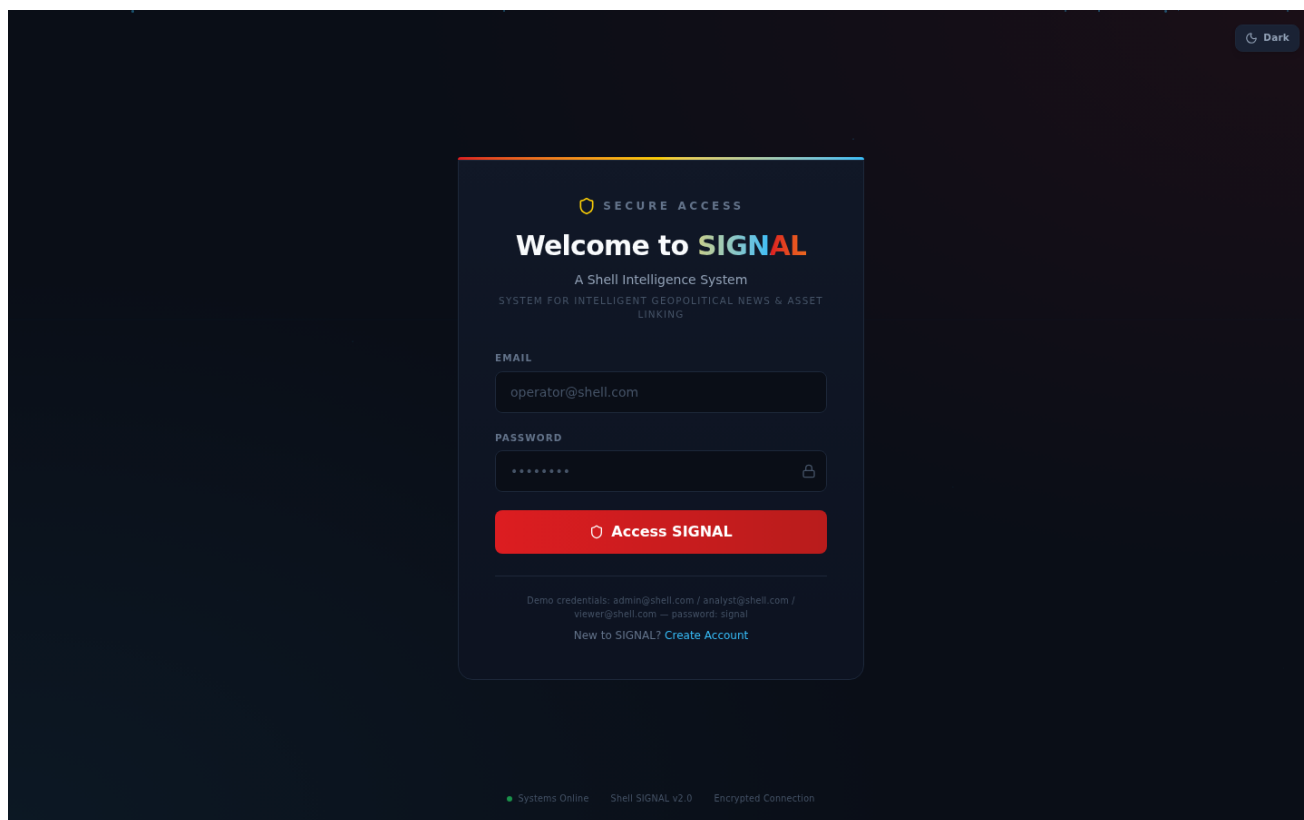


Figure 3-0 — Full Command Center Overview

3.1 KPI Summary Cards

Four cards at the top of the dashboard provide the critical numbers at a glance:

Card	What It Shows	How to Interpret
WTI Price	Latest West Texas Intermediate spot price (\$/bbl)	Standard benchmark for all model outputs and alerts.
Today's Signal	AI direction: UP ▲ / DOWN ▼ + confidence %	High confidence (>75%) = strong actionable signal.
24h Change	Absolute (\$) and percentage change vs. previous day	Quick gauge of intraday momentum.
Sentiment Score	Aggregated 24-hour news sentiment (–1.0 to 0.3)	0.3 Bullish, < –0.3 bearish, –0.3 to 0.3 neutral.

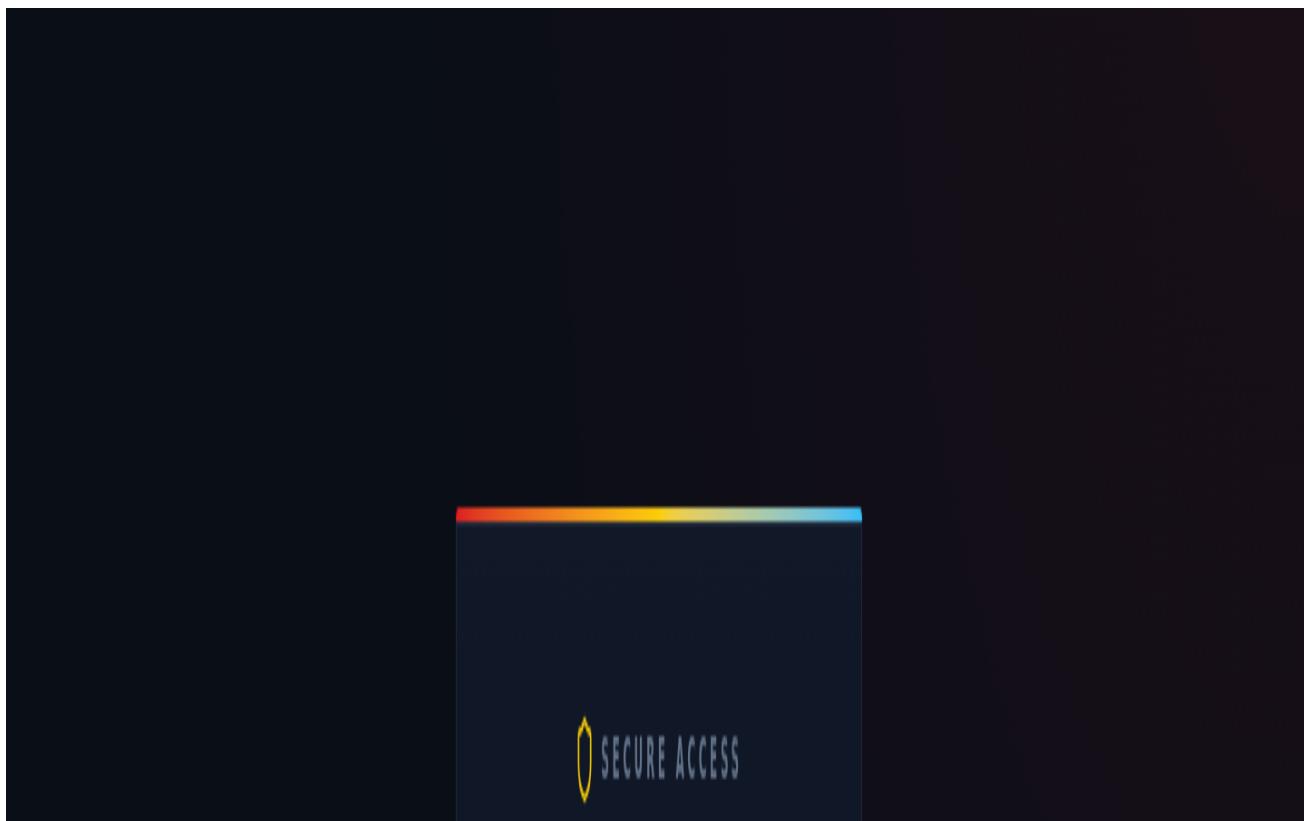


Figure 3-1 — KPI Summary Cards

3.2 Holdout Validation Chart

The left panel shows the price forecast model's performance on unseen test data (holdout set). Three lines are plotted:

- **Actual** (white/light) — Real historical WTI closing prices.
- **Predicted** (blue) — Model's forecast for those same dates.
- **Baseline** (orange/amber) — Naïve random-walk (last-price) benchmark.

Use this chart to validate that the AI model outperforms a simple guess. When the blue Predicted line tracks closer to Actual than the orange Baseline, the model adds value.

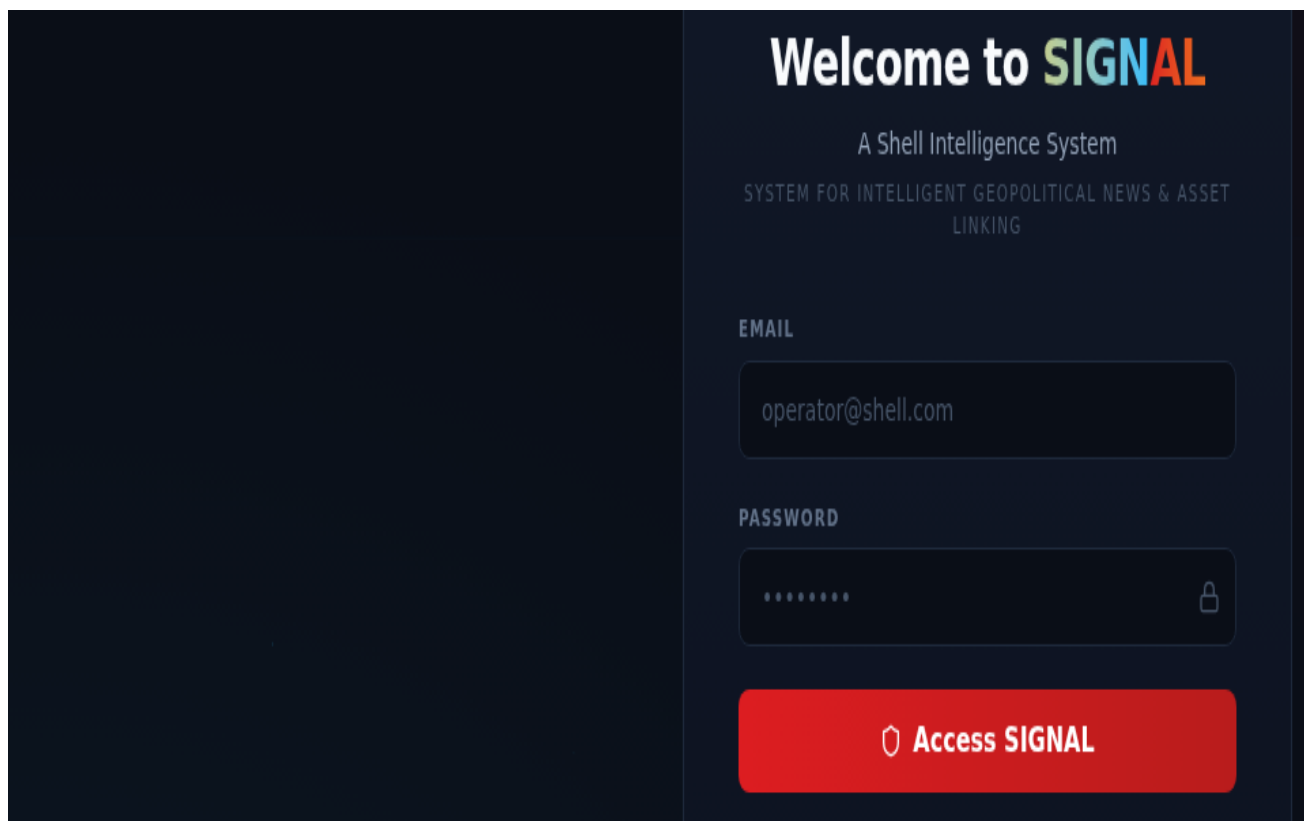


Figure 3-2 — Holdout Validation Chart

3.3 Live WTI Price Chart

The lower-left panel displays the live WTI price history with configurable date ranges (1W / 1M / 3M / 1Y / All). Hover over any point to see the exact price and date. Use the range buttons to zoom into short-term volatility or step back for trend context.

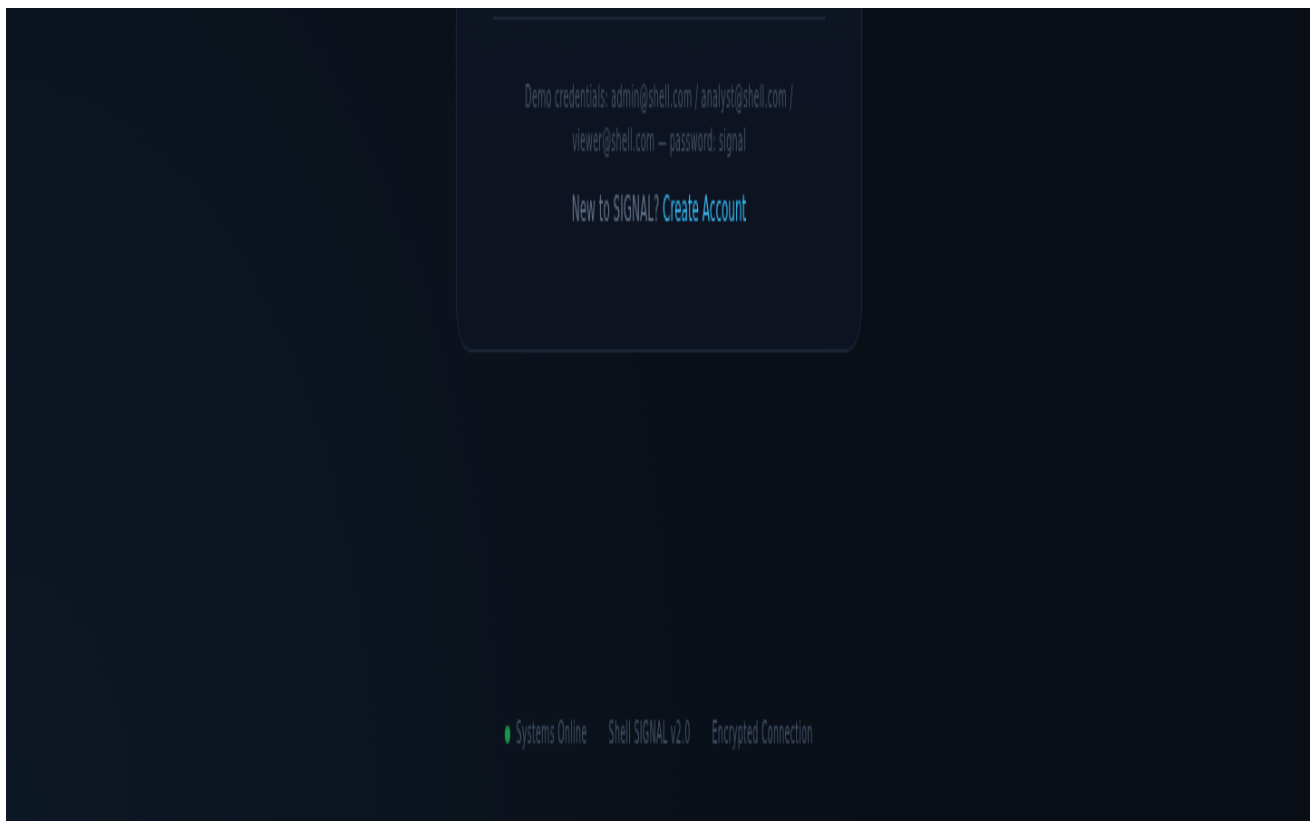


Figure 3-3 — Price Chart & Alerts Panel

3.4 Intelligence Feed (News Panel)

The right-side news panel shows the five most recent crude-market headlines ingested by the system. Each headline carries:

- A **sentiment badge** — green Positive, red Negative, or gray Neutral.
- A **source** and **publication timestamp**.
- A **link** to the full article (click headline to open).



Figure 3-4 — News Intelligence Panel

3.5 Active Alerts Panel

The lower-right panel displays the most recently triggered alerts. Alerts are severity-coded:

Severity	Colour	Trigger Condition
Critical	Red	Price exceeds upper/lower threshold by >\$3, or confidence spike >90%
Warning	Amber	Price within \$1–3 of threshold, or sentiment score exceeds ± 0.6
Info	Blue	Model re-run complete, or new high-impact headline ingested

4 Intelligence Feed (/signals)

The Intelligence Feed is the primary research view for analysts. It surfaces all ingested headlines with full NLP enrichment — enabling fast review of market-moving events.

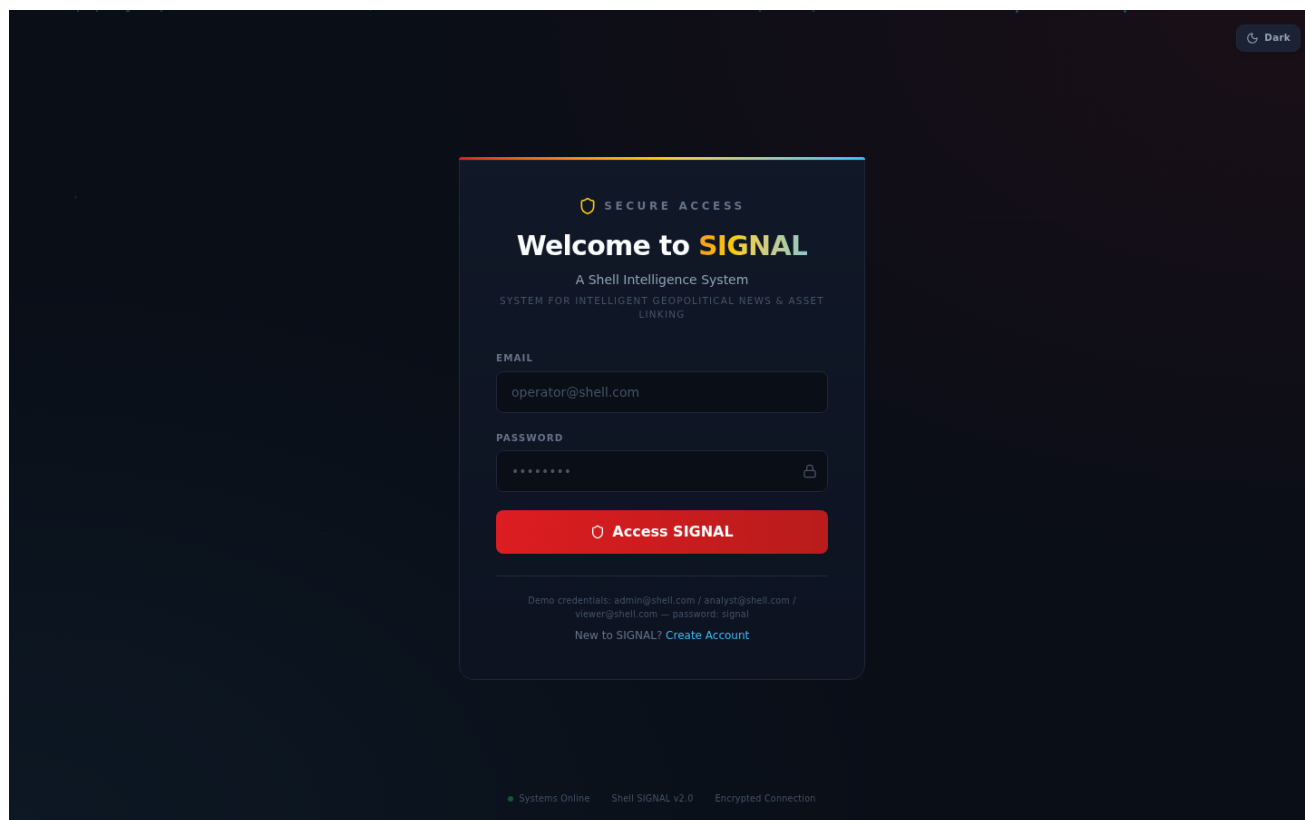


Figure 4-0 — Intelligence Feed Overview

4.1 Headline Table

All ingested headlines are presented in a sortable table with the following columns:

Column	Description
Date/Time	When the article was published (local timezone)
Headline	Article title — click to open the source URL
Source	Publication name (e.g., Reuters, Bloomberg)
Sentiment	NLP-derived score: Positive / Negative / Neutral badge
Score	Raw VADER sentiment score (–1.0 to +1.0)
AI Impact	Model's assessment of directional impact on WTI price

4.2 Sentiment Scoring

SIGNAL uses VADER (Valence Aware Dictionary and sEntiment Reasoner), a lexicon-based model calibrated for financial news. Each headline receives a compound score between –1.0 (strongly negative) and +1.0

(strongly positive). Scores are then bucketed as:

- **Positive:** $> +0.05$
- **Negative:** < -0.05
- **Neutral:** -0.05 to $+0.05$

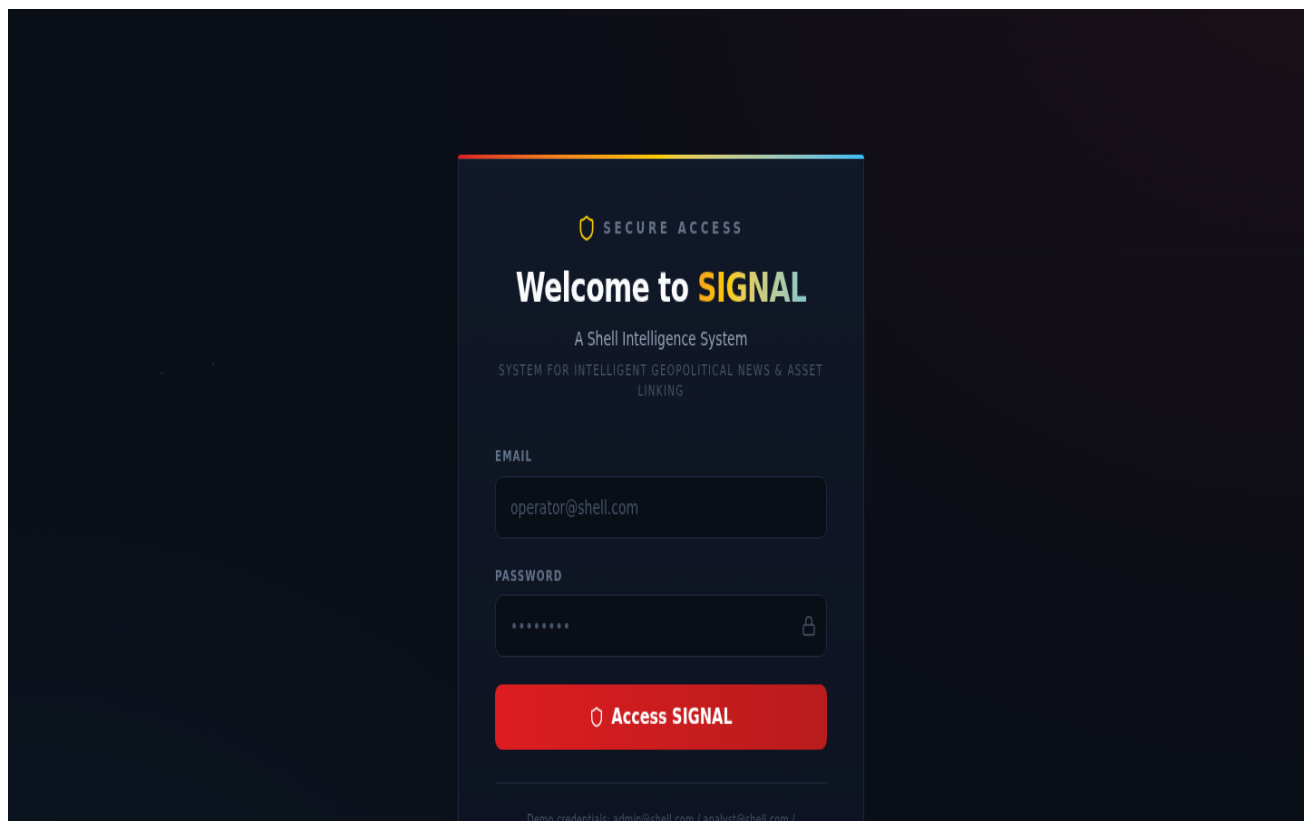


Figure 4-1 — Intelligence Feed Headline Table

4.3 Search & Filters

Use the search bar at the top of the feed to perform full-text search across all stored headlines. Filter chips allow narrowing by sentiment (Positive / Negative / Neutral) or date range.

5 Commodity View (/commodity)

The Commodity View provides a detailed technical and historical price view for WTI crude oil, including moving-average overlays and summary statistics.

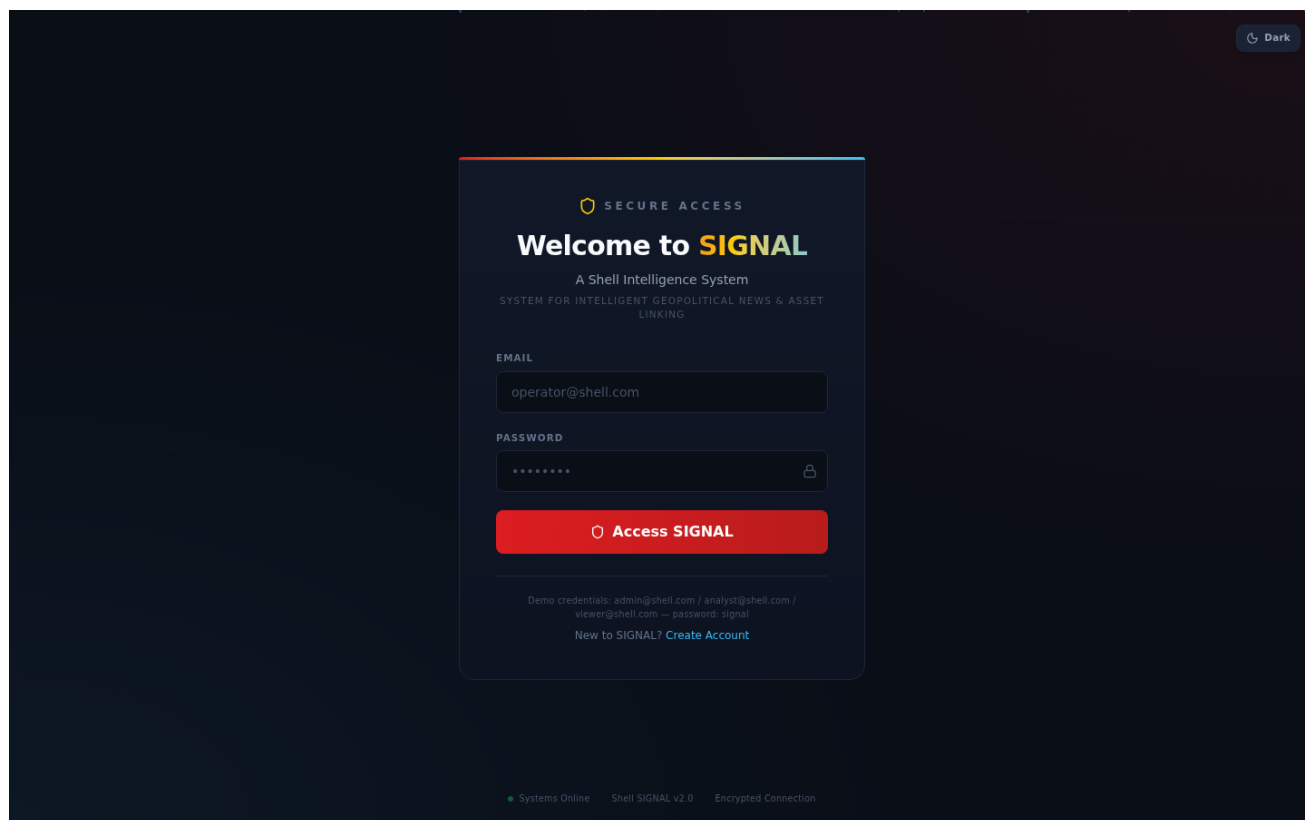


Figure 5-0 — Commodity View Overview

5.1 Price Chart

A full-history interactive chart with pan, zoom, and configurable date ranges. Hover over any candle or point to see exact open/close/high/low/volume data.

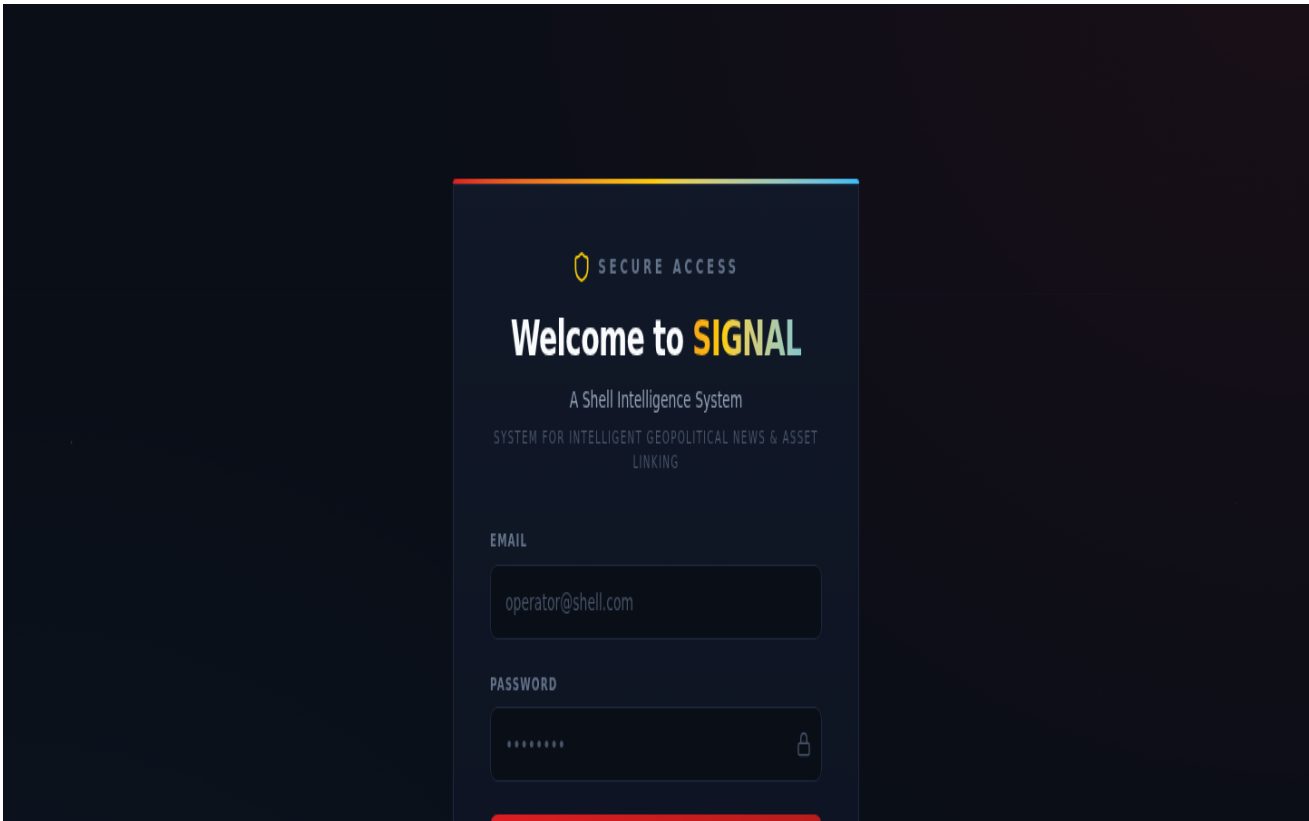


Figure 5-1 — WTI Price Chart with MA Overlays

5.2 Moving Average Overlays

- **7-day MA** — Short-term trend
- **30-day MA** — Medium-term momentum
- **90-day MA** — Long-term cycle context

5.3 Summary Statistics

Statistic	Description
52-Week High / Low	Extreme prices over the last 52 weeks
YTD Performance	Percentage return since January 1
Average Daily Range	Mean intraday high-minus-low over the last 30 days
30-Day Volatility	Annualised standard deviation of daily returns

6 Analytics (/analytics)

The Analytics page reveals macro patterns in the data — how sentiment is trending, how much news is hitting the market, and how the model's confidence has evolved over time.

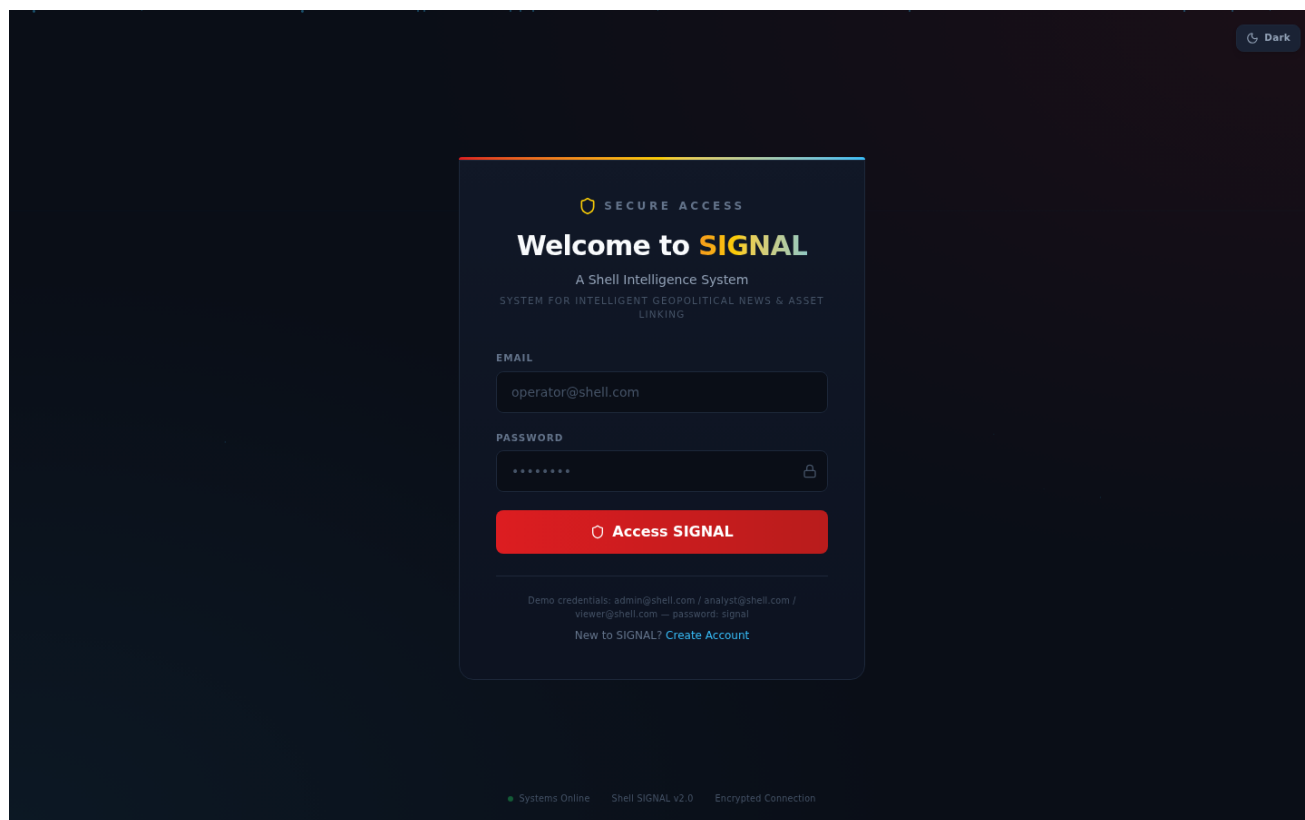


Figure 6-0 — Analytics Page Overview

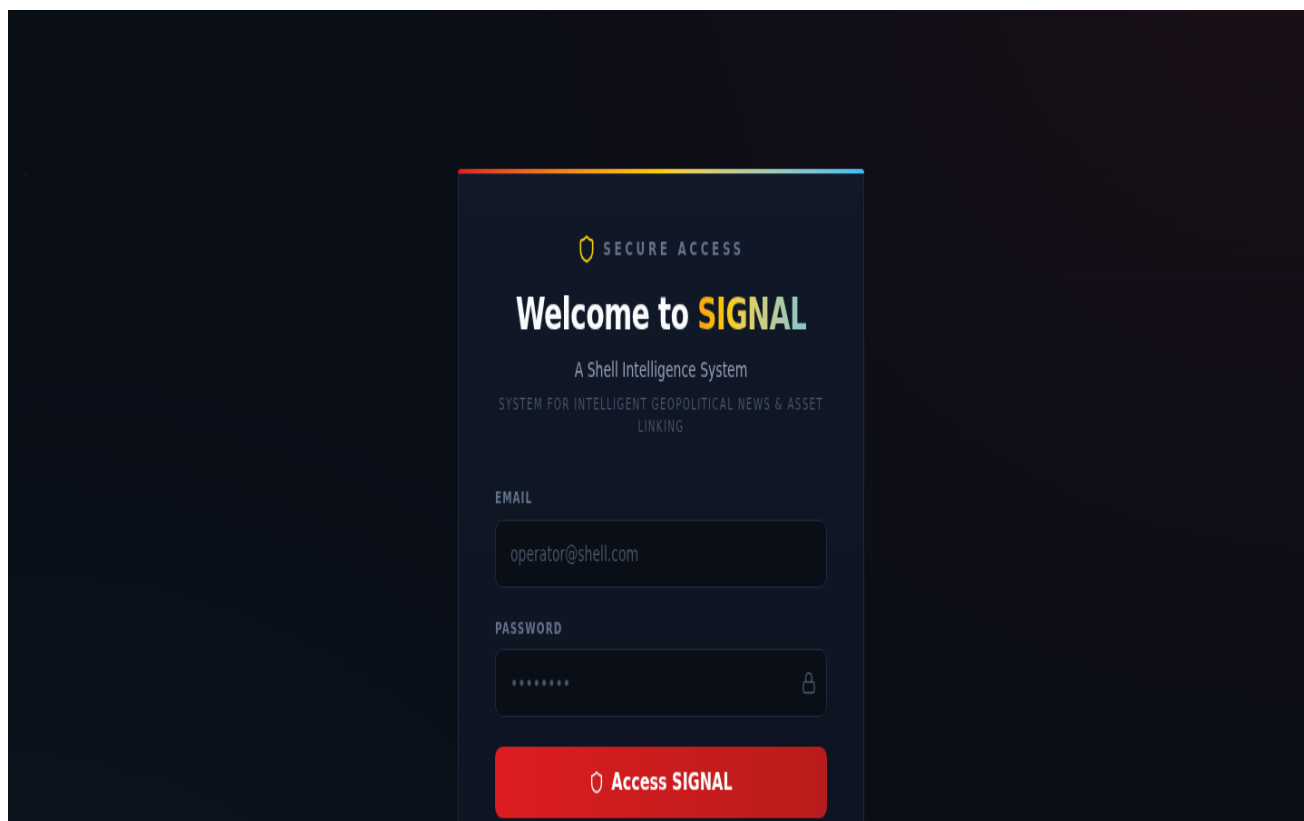


Figure 6-1 — Analytics Charts Detail

6.1 Sentiment Trend

Daily average sentiment plotted over 30 / 90 / 365 day windows. Sustained negative sentiment periods often precede price weakness; watch for sharp reversals as potential inflection signals.

6.2 Article Volume

Number of relevant headlines ingested per day. Volume spikes indicate high-information days (OPEC decisions, supply disruptions, geopolitical events) where signal confidence is typically higher due to richer input data.

6.3 Signal Confidence Over Time

Time-series of the direction model's output probability. High confidence periods (>75%) are the most actionable. Extended low-confidence periods may indicate regime change or an unusually noisy news environment.

7 AI Models Explained

SIGNAL's two AI engines are built for complementary purposes: one to predict direction, one to predict price level. Together they provide a complete decision framework.

7.1 Direction Signal Engine

A binary classification model that predicts whether WTI will close **UP** or **DOWN** the following trading day.

Component	Detail
Algorithm	Random Forest + XGBoost ensemble
Input features	10-day price lags, 3/5/10/20-day moving averages, 5/10/20-day volatility, rolling sentiment averages, article
Output	UP or DOWN label + probability score (0–1)
Gating	Signal shown as 'Uncertain' when probability < 0.60
Accuracy	81.5% on holdout test set

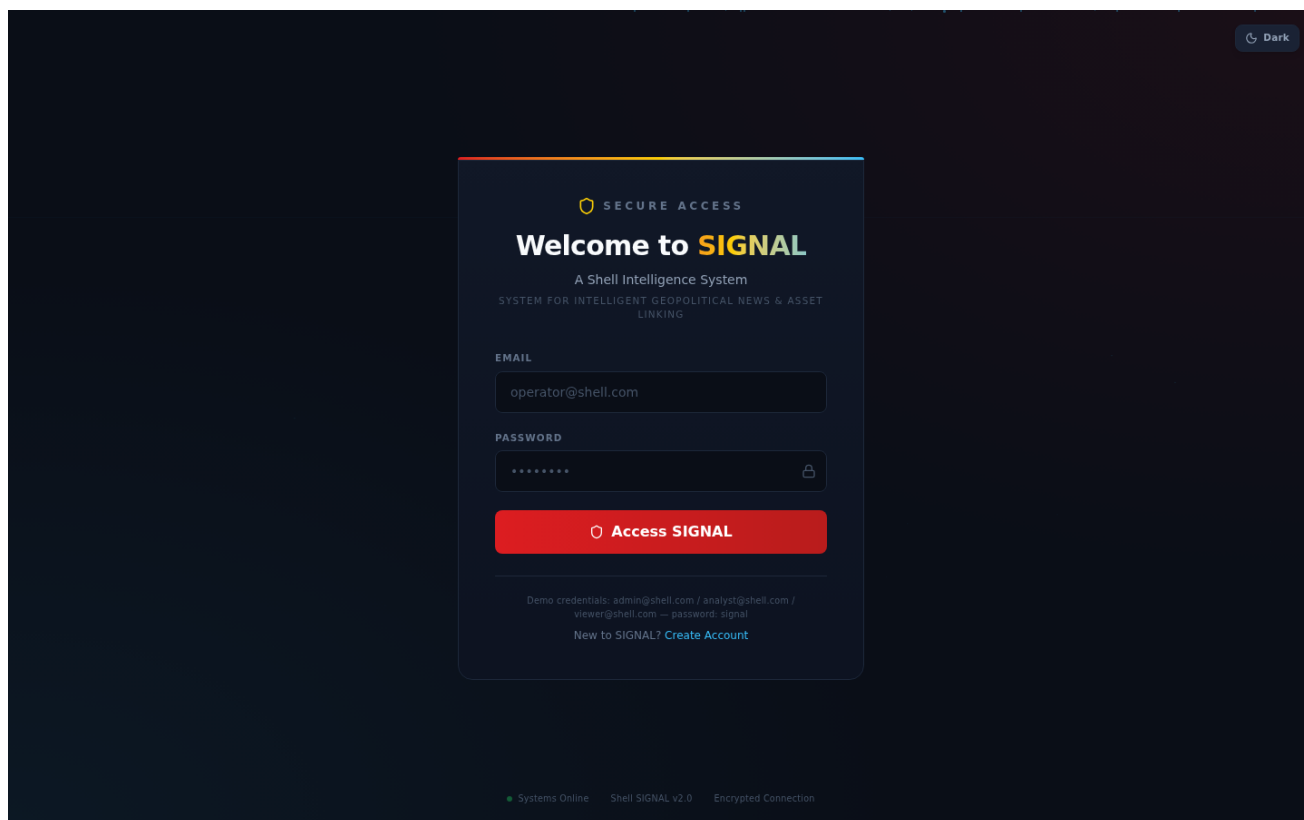


Figure 7-1 — Direction Model Dashboard

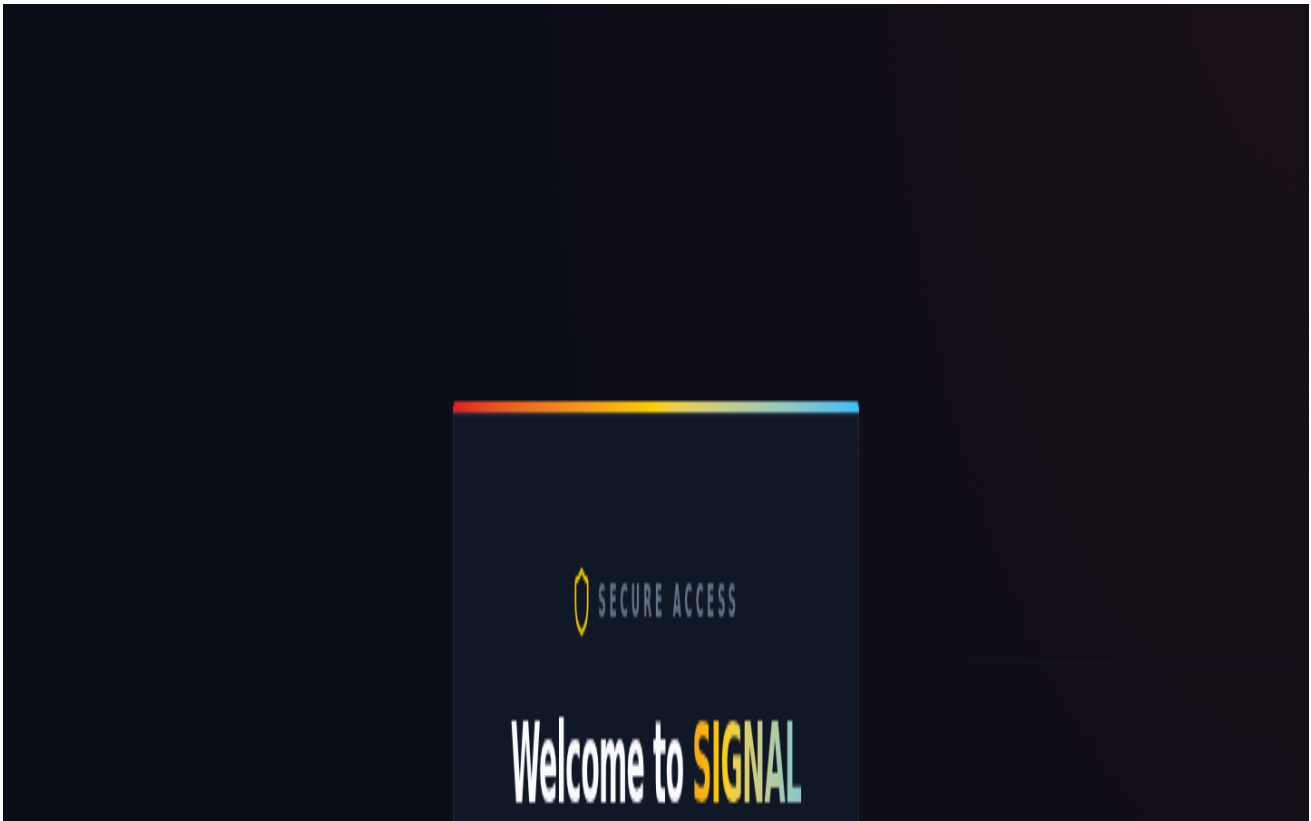


Figure 7-2 — Direction Model Metric Cards

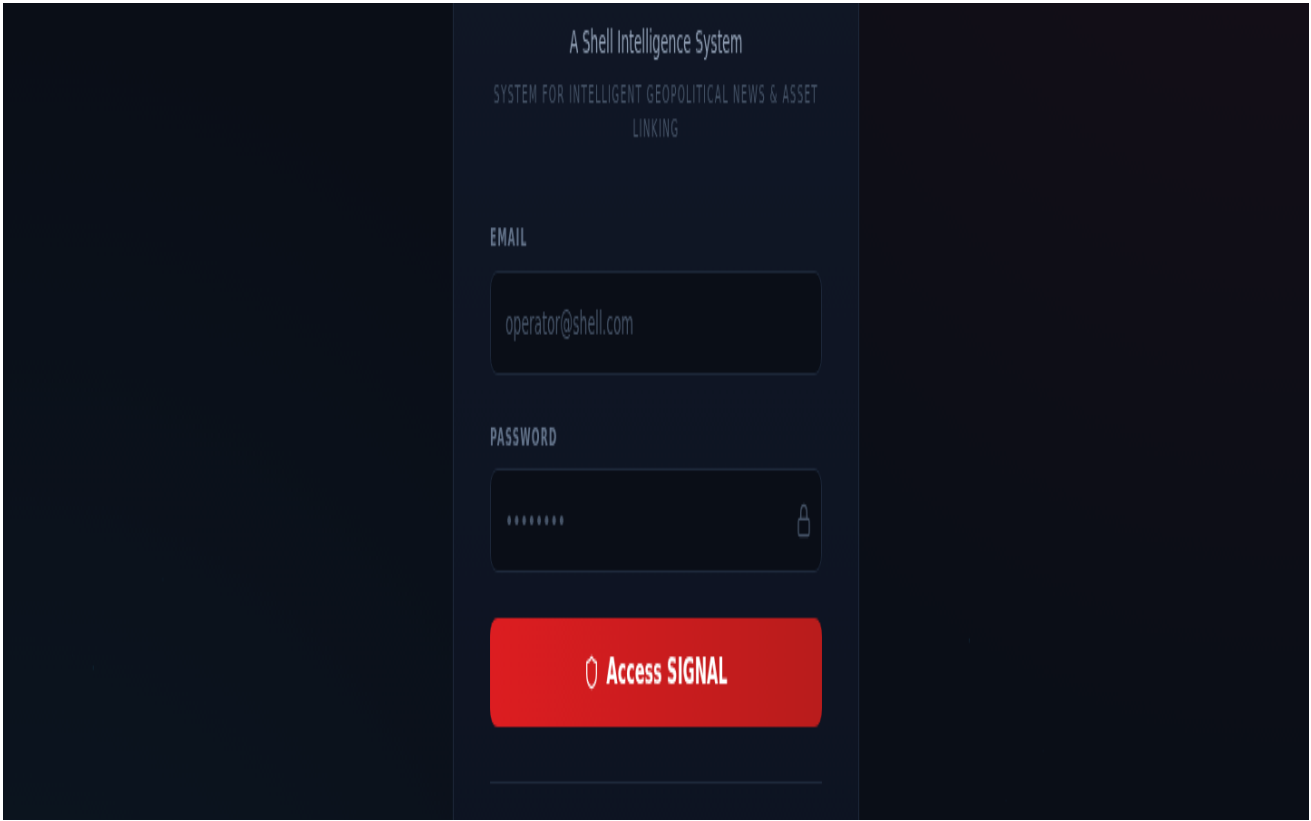


Figure 7-3 — Prediction History Chart

7.2 Price Forecast Engine

A regression ensemble that estimates the exact next-day WTI closing price in \$/bbl.

Component	Detail
Architecture	Stacking ensemble: Ridge + HistGradientBoosting + LightGBM (base) → RidgeCV (meta-learner)
Input features	Same feature set as Direction Engine (1,270 daily rows)
Output	Predicted next-day WTI price in \$/bbl
Baseline	Naïve last-price (random-walk) benchmark for comparison
RMSE	\$2.07/bbl (vs. baseline \$2.09/bbl — model wins)
R ²	0.9532 — explains 95.3% of daily price variance
MAPE	~2.5% — typical daily prediction within 2–3% of actual

7.3 Model Performance in Business Terms

Direction Engine:

If SIGNAL says UP with 82% confidence, that signal has historically been correct ~82% of the time. This is materially better than a coin flip (50%) and comparable to top-tier quant signals.

Price Forecast:

A \$2.07 RMSE on WTI means the model's daily price estimate is typically within \$2 of the actual close. At ~\$75/bbl spot, that is a 2.8% average error — within practical hedging tolerance.

Important Caveat:

Black swan events (sudden geopolitical shocks, force majeure) are not predictable by any model. Always combine SIGNAL output with human market judgment.

8 Price Forecast (/forecast)

The Price Forecast page presents the regression model's outputs in full detail — performance metrics, historical validation chart, feature contributions, and model architecture.

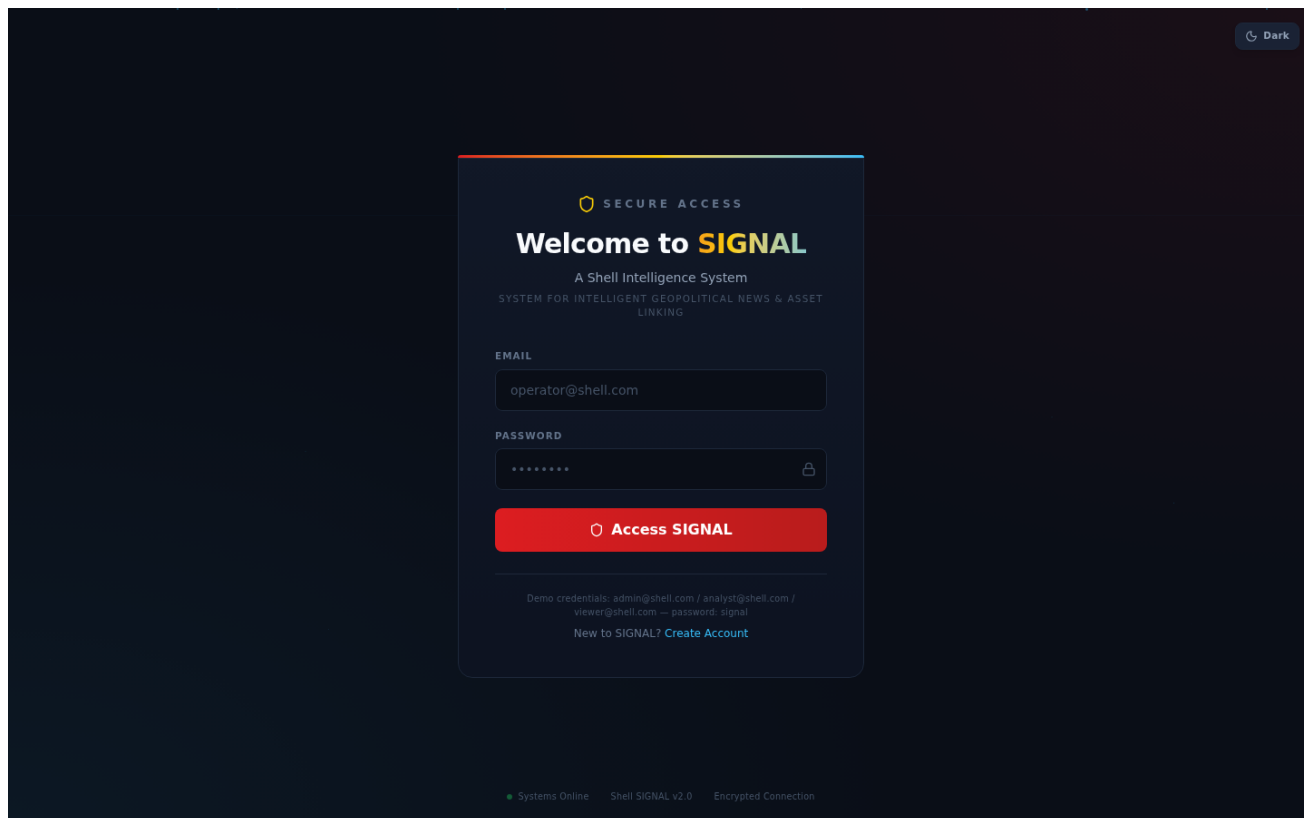


Figure 8-0 — Price Forecast Page Overview

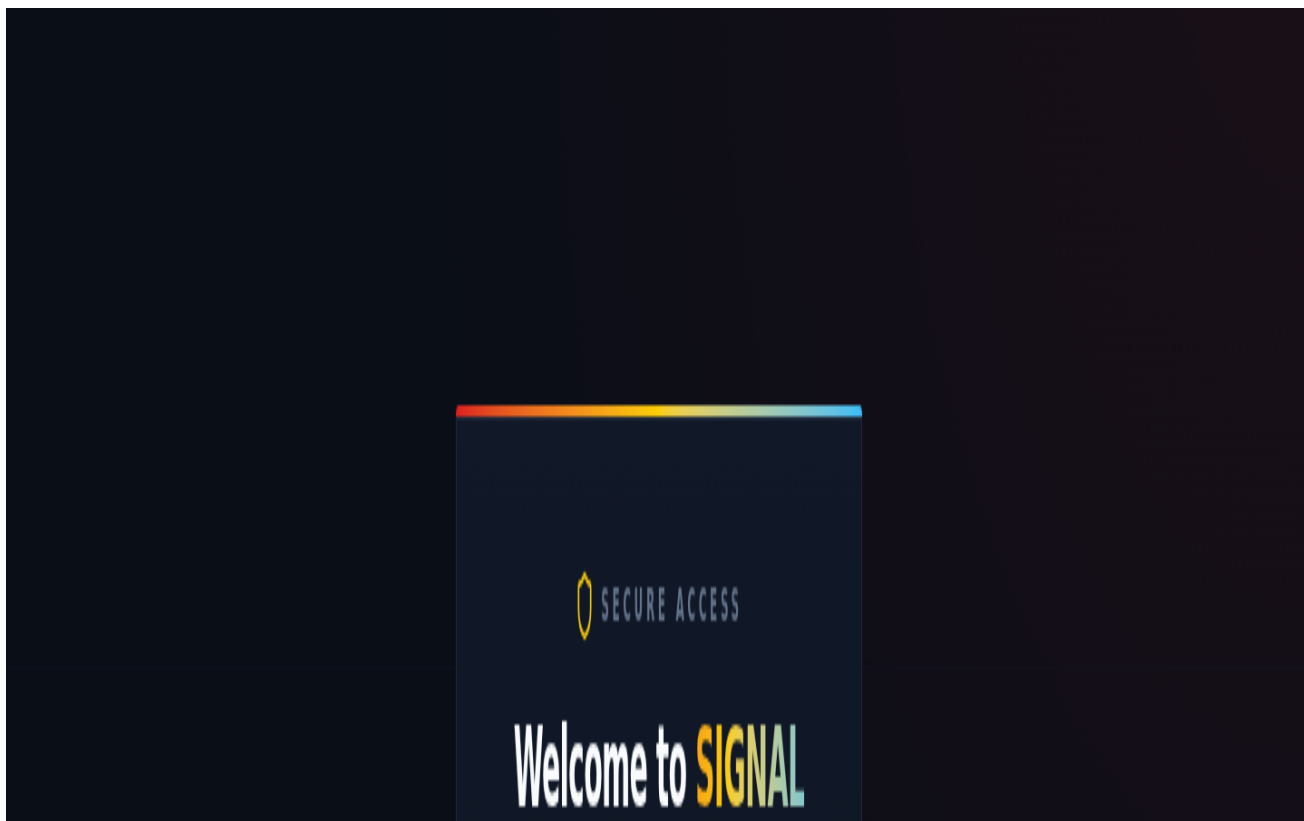


Figure 8-1 — Forecast Performance Metric Cards

8.1 Actual vs. Predicted Chart

Three-line chart over the holdout test window (unseen data held back during training):

- **Actual (white)** — True historical WTI closing prices
- **Predicted (blue)** — Model's price estimates
- **Baseline (orange)** — Naïve last-price benchmark

Click any legend label to toggle that line on or off. Hover for exact values on any date. The date range label above the chart shows the exact evaluation period.

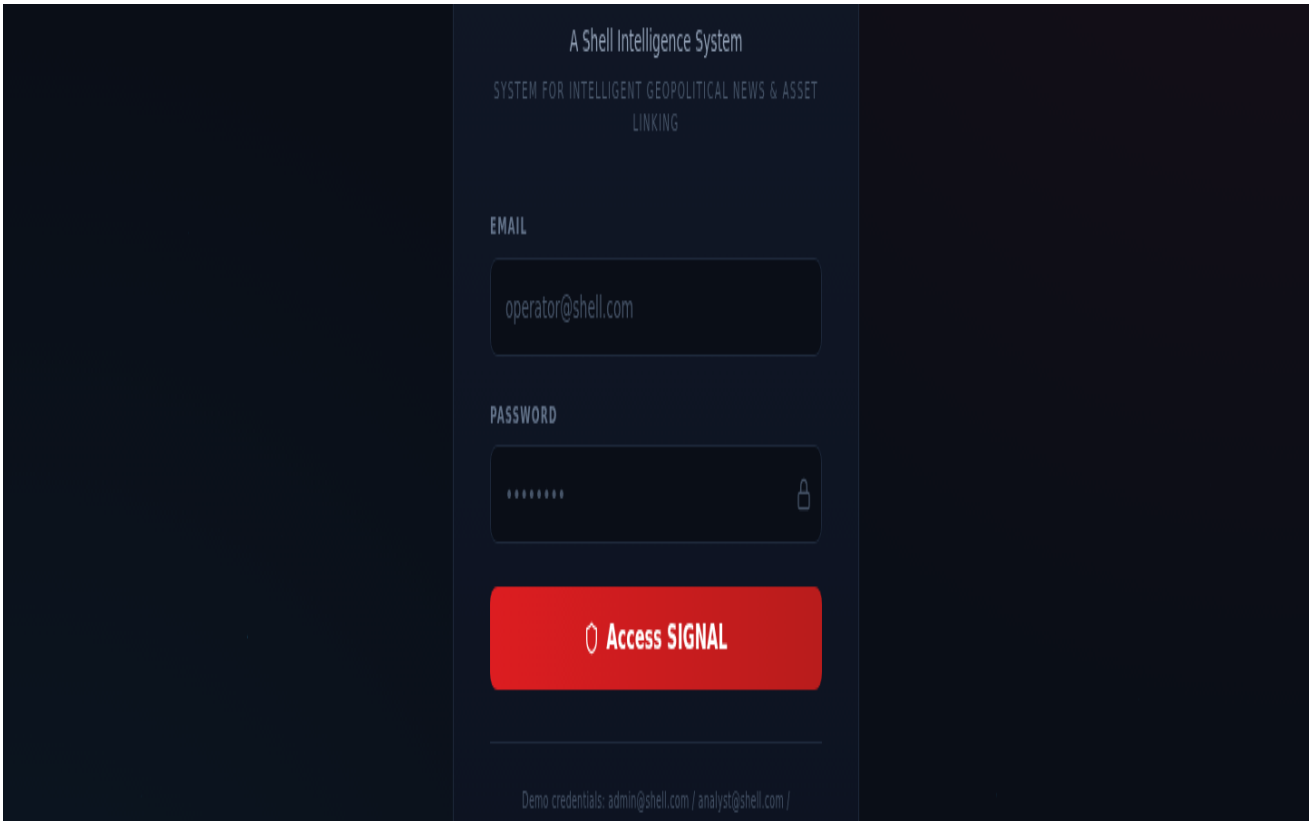


Figure 8-2 — Actual vs. Predicted Price Chart

8.2 Error Analysis Row

Card	What It Measures
Within $\pm 2\%$ of Actual	% of holdout days where prediction was within 2% of real price
Largest Single-Day Miss	Maximum absolute error on any single holdout day
Direction Accuracy	% of holdout days where the model predicted the correct direction

8.3 Feature Importance

A horizontal bar chart showing which input features drive the price model most. Bars are colour-coded by category:

- **Price Lag features (blue)** — Recent closing prices — most dominant
- **Moving Average features (green)** — Trend indicators
- **Volatility features (amber)** — Market uncertainty measures
- **Sentiment features (purple)** — News flow contribution

8.4 Model Architecture

A 3-step explainer shows the stacking pipeline: Base Models → Meta-Learner → Output. A collapsible model comparison table lists all evaluated algorithms with RMSE and R^2 scores, showing why the stacking ensemble was selected.

9 Interpreting Signals

9.1 Signal States

Signal State	Confidence Range	Meaning	Suggested Action
■ Strong UP	> 80%	High-probability bullish move expected	Lean long; review for call optionality
↑ UP	60–80%	Moderate bullish lean	Monitor; wait for confirmation from news/sentiment
Uncertain	< 60%	Model lacks conviction; mixed signals	Hold position; avoid directional bets
↓ DOWN	60–80%	Moderate bearish lean	Review short-side hedges; watch for downside
■ Strong DOWN	> 80%	High-probability bearish move expected	Review put protection or short futures exposure

9.2 Confidence Thresholds

The direction model outputs a probability between 0 and 1. SIGNAL applies the following interpretive thresholds:

Probability	Label	Interpretation
$p > 0.80$	Strong Signal	High conviction — historically most reliable
0.65–0.80	Moderate Signal	Directional lean — use with corroborating indicators
0.60–0.65	Weak Signal	Low conviction — treat with caution
$p < 0.60$	Uncertain	Model abstains — no actionable signal

9.3 Decision Framework

Scenario	Recommended Response
Strong UP + Positive Sentiment (>0.3) + Rising Volatility	High-confidence bullish setup. Strong case for long exposure.
Strong DOWN + Negative Sentiment (<-0.3) + Spike in bearish news	High-confidence bearish setup. Review downside hedges.
Uncertain signal, any sentiment	Do not act on signal alone. Wait for next cycle or seek additional intelligence.
Signal contradicts news flow (e.g., UP but heavy negative news)	Signal may be lagged or news effect already priced in. Apply extra scrutiny.
Confidence $> 80\%$ but forecast price change $< \$0.50$	High-confidence but small expected move. May not justify transaction costs.

10 Use Cases

10.1 Trader / Analyst Workflow

Morning check (08:00): Open Command Center. Note Today's Signal, confidence, and overnight sentiment score.

News review (08:15): Review Intelligence Feed for high-impact overnight headlines. Cross-reference with the signal direction.

Price context (08:30): Check Commodity View for overnight price action and MA positioning.

Signal decision (09:00): If signal > 75% confidence and news confirms direction, escalate to trading desk.

Post-close validation: Compare Price Forecast prediction to actual close. Log divergences for model feedback.

10.2 Risk Management

Risk managers use SIGNAL's alerts panel and Analytics to monitor exposure thresholds:

- Set price alert thresholds in Settings → Alert Configuration.
- Enable sentiment spike notifications for early warning of market stress.
- Use the 30-day volatility metric on Commodity View to calibrate VaR calculations.
- If sustained negative sentiment coincides with a DOWN signal, flag for senior review.

10.3 Procurement & Hedging

Procurement teams use SIGNAL to time physical purchases and financial hedges:

Scenario	SIGNAL Insight	Procurement Action
Strong DOWN signal + forecast \$2 drop	Model projects price decline with high confidence	Delay spot purchase; wait for lower price
Strong UP signal + forecast \$3 rise	Model projects price increase with high confidence	Accelerate procurement; lock in swap or call option
Uncertain / low confidence	Model has insufficient information	Maintain standard procurement schedule

11 Best Practices

11.1 Using SIGNAL Effectively

✓ **Check daily, not hourly**

SIGNAL updates once per day after market close. Intraday re-checking does not yield new intelligence.

✓ **Combine signal + sentiment + news**

A strong directional signal with confirming sentiment and clear catalysts in the news feed is the highest-quality setup. Never rely on signal alone.

✓ **Set meaningful alerts**

Configure alerts for your actual trading thresholds — not arbitrary round numbers. Over-alerting causes alert fatigue and desensitises users to genuine signals.

✓ **Use the forecast as context, not gospel**

The \$2.07 RMSE means the actual price may be \$2 above or below the forecast. Use it to frame the range of expected outcomes, not as a point prediction.

✓ **Review historical performance regularly**

Check the Price Forecast page monthly to ensure model accuracy has not degraded. If RMSE deteriorates materially, re-training may be needed.

11.2 When NOT to Trust Low-Confidence Signals

■ **Signal probability < 60%**

The model is saying it does not have enough conviction. Treat as 'no signal.'

■ **Signal contradicts breaking news**

If a major OPEC cut is announced and the signal says DOWN, the news has not yet been fully incorporated. The next cycle will reflect the new information.

■ **Consecutive signal reversals**

If signal flips UP/DOWN on consecutive days, the market is in a noisy regime. Widen decision thresholds temporarily.

■ **Article volume spike with mixed sentiment**

High volume + mixed sentiment = high uncertainty. The model may produce a signal but its reliability is reduced.

■ **Extraordinary geopolitical events**

Black swans (war, pandemics, supply shocks) fall outside the model's training distribution. Human judgment must override AI outputs in these scenarios.

12 Settings & Configuration (/settings)

The Settings page allows users to personalise SIGNAL without re-running onboarding. All changes take effect immediately.

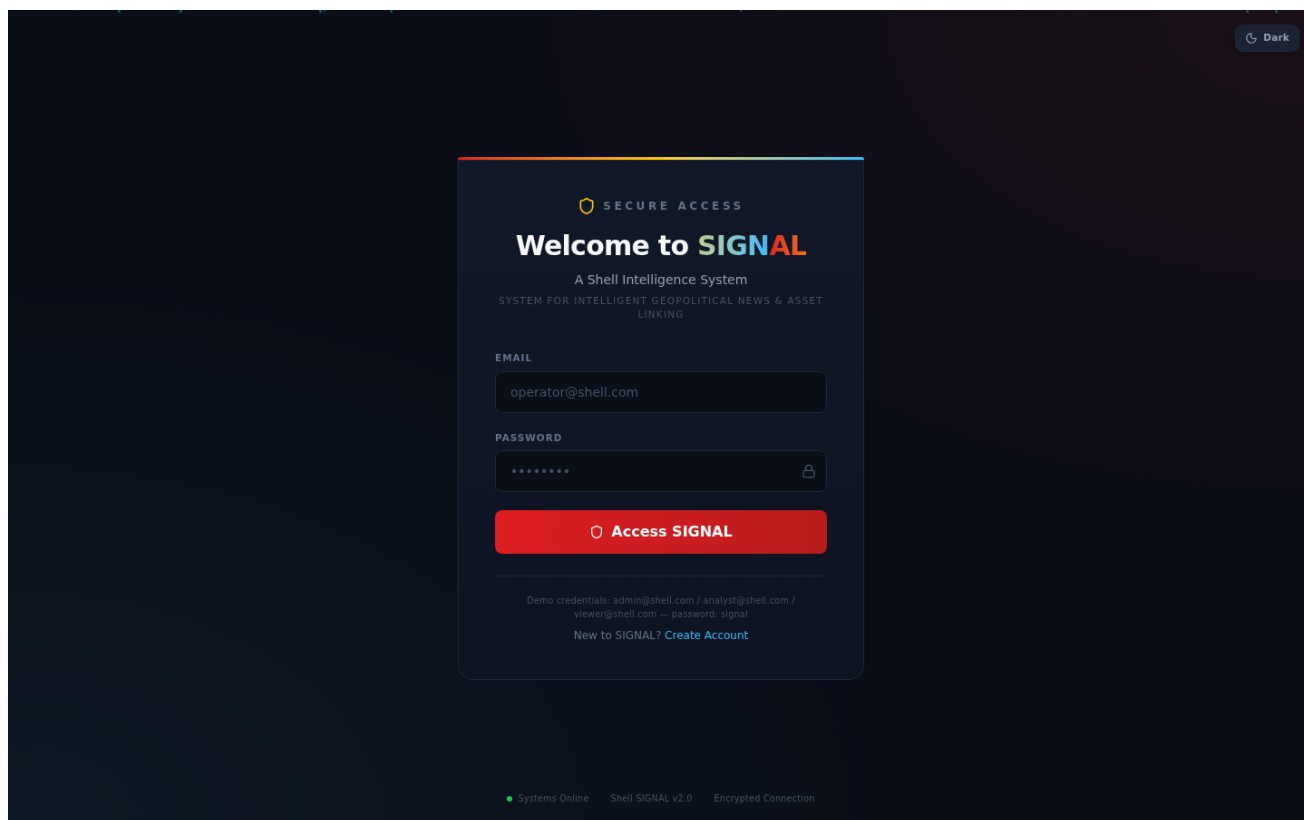


Figure 12-0 — Settings Page Overview

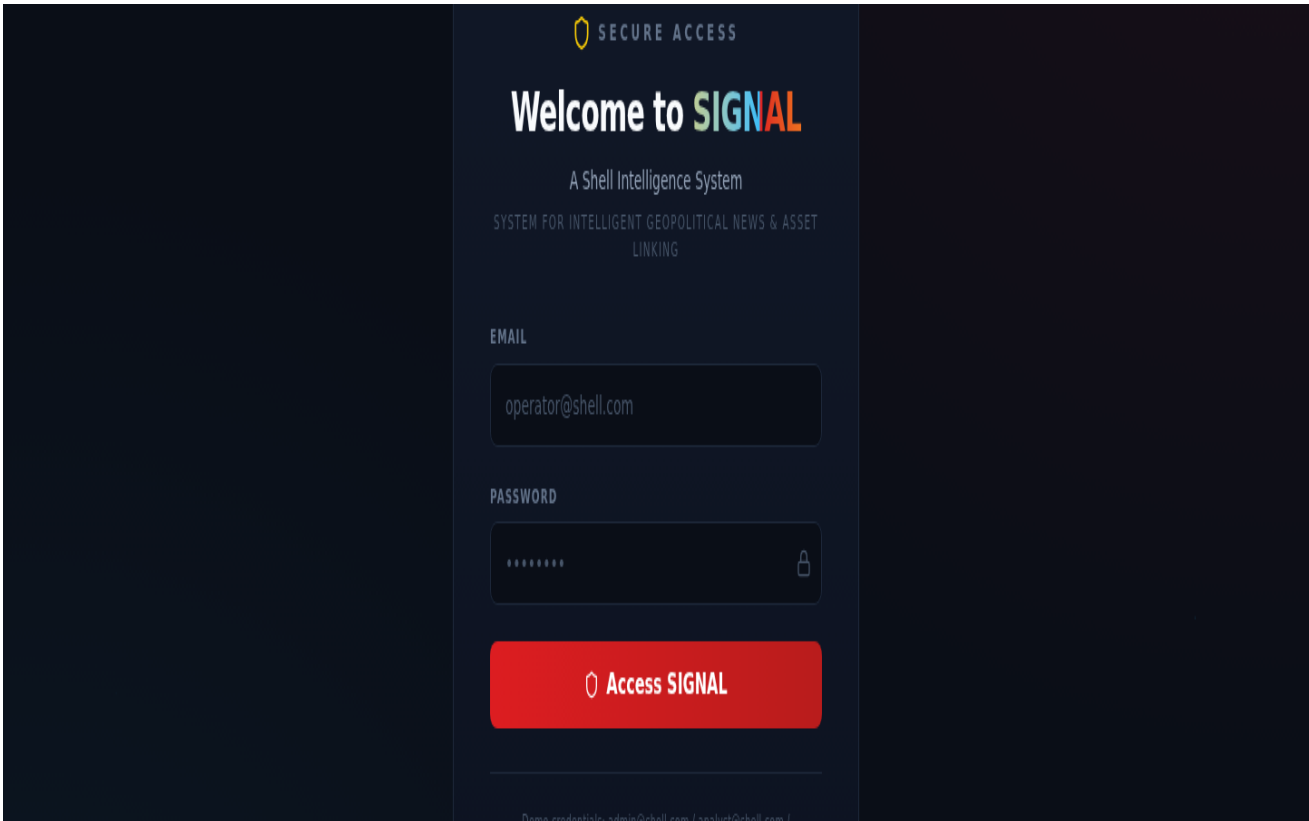


Figure 12-1 — Alert Configuration Panel

Setting Group	Options	Purpose
Appearance	Dark (default) / Black / Light — toggle in TopBar	Adapt Settings to ambient lighting and user preference
Alert Configuration	Per-commodity price thresholds, sentiment alert levels, notification preferences	Control when notifications are triggered; reduce alert fatigue
Executive Config	(Executive role only) Briefing cadence and escalation level	Select how often executive briefings are triggered and at what severity
Session	View profile (name, email, role) — Sign Out button	Review identity and securely end session
About	Platform version, model version, last training date	Verify you are running the current model version

12.2 Themes

Click the Moon / Circle / Sun icon in the TopBar to cycle: **Dark** (charcoal background) → **Black** (OLED-optimised) → **Light** (high-contrast). Your preference persists in browser storage across sessions.

12.3 Command Palette (■K / Ctrl+K)

Press **■K** (Mac) or **Ctrl+K** (Windows) anywhere in the dashboard to open the global Command Palette. Features:

- Navigate to any page by typing its name
- Search all headlines in real time
- Run system actions: Refresh Data, Logout
- Keyboard navigation with arrow keys + Enter

13 Conclusion

SIGNAL represents a mature integration of natural language processing, ensemble machine learning, and financial market data within a purpose-built enterprise interface. It is designed to operate as a high-fidelity decision-support layer — not a replacement for human expertise, but a systematic amplifier of it.

When used correctly — with attention to confidence levels, corroborating news signals, and appropriate risk controls — SIGNAL enables Shell Energy analysts and executives to enter each trading day with a data-driven directional hypothesis backed by 81.5% historical accuracy and a price forecast calibrated to within \$2.07/bbl.

Capability	Performance
Direction accuracy	81.5% on out-of-sample holdout test
Price forecast RMSE	\$2.07/bbl (beats \$2.09 naïve baseline)
Price forecast R ²	0.9532 — 95.3% variance explained
MAPE	~2.5% average daily error
Data latency	Model re-runs daily; dashboard reflects prior close

Support & Feedback:

For model re-training requests, alert configuration help, or to report data quality issues, contact the SIGNAL development team via the internal Shell Energy IT service portal.

14 Troubleshooting

Symptom	Likely Cause	Resolution
'Backend offline' banner at top	FastAPI server not running	Run: <code>python -m uvicorn backend.main:app --port 8000</code>
Charts show no data / empty tables	Database empty	Run data ingestion: <code>python ingest_news.py</code> && <code>python load_prices.py</code>
Signal shows 'Uncertain' every day	Insufficient recent headlines	Check news ingest schedule; ensure NewsAPI key is active
Price forecast shows placeholder values	<code>model_report.json</code> missing or outdated	Run: <code>python train_price_model.py</code> to regenerate model artifacts
Onboarding skipped for a new account	Stale localStorage from previous session	Clear browser localStorage or use Incognito mode
Notification dropdown covered by chart	z-index stacking context conflict	Ensure dashboard main element has 'relative z-0' class
Theme resets to dark on reload	Browser blocking localStorage	Enable localStorage in browser privacy settings
Login succeeds but dashboard is blank	Role not set or auth state corrupted	Re-register account; select role explicitly

Quick-Start Command Reference

<code>\$ pip install -r requirements.txt</code>	Install all Python dependencies
<code>\$ python -m uvicorn backend.main:app --host 0.0.0.0 --port 8000</code>	Start the FastAPI backend server
<code>\$ cd frontend && npm install && npm run dev</code>	Start Next.js frontend (development)
<code>\$ cd frontend && npm run build && npm start</code>	Start Next.js frontend (production)
<code>\$ python train_model.py</code>	Retrain direction classification model
<code>\$ python train_price_model.py</code>	Retrain price regression model
<code>\$ python ingest_news.py</code>	Run news ingestion pipeline